lecture 2

asymptotic notation

big O 1 unit of time per instruction to define cost

```
def selection_sort(A,N):
    for i ∈ 0,1,...,N-1:
        m ∈ i
        for j ∈ i+1,....,N-1:
            if A[m] > A[j]:
                m ∈ j
        swap A[i] ↔ A[m]
    return A
```

destroy cost: $\frac{1}{2}N(N-1)$

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2 big things with chart

1. 1 unit of time → strong assumption
   - some need compare, other need load/store. different types of instructions
2. hard to analyze
   need an easier way to analyze

Big O asymptotic analysis
- ignore constant factors all instructions are O(1)
- look at leading term $\frac{1}{2}N(N-1)+1 \rightarrow O(N^2)$
  $\frac{1}{2}N^2 - \frac{1}{2}N + 1$ ignore $-\frac{1}{2}N+1$

Def:

$$O(f(n)) = \{ f_0(n) \mid \exists C, N, \forall n \geq N,\ 0 \leq f_0(n) \leq C f(n) \}$$

$O(f(n))$ - define a set of functions that all grow no faster than $f(n)$

$g(n) \in O(f(n))$

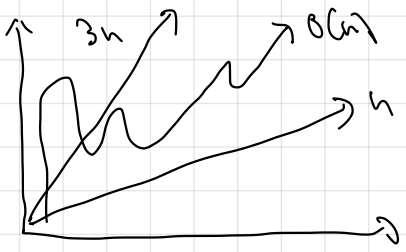
$f_0(n)$ is a candidate function being tested in that set

| such that
∃ there exists
$\forall n \geq N$ for all n greater or equal to N $0 \leq f_0(n) \leq C f(n)$

condition: $f_0(n) \in O(f(n))$ if $f_0(n)$ is eventually upper bounded by $f(n)$

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ex: $O(n) = \{ f_0(n) \mid \exists C, N, \forall n \geq N,\ 0 \leq f_0(n) \leq Cn \}$
$= \{ 5n, n+1, n, 6n+2, \dots \}$



okay if the function is greater than the upper bound at the beginning

how is this O(n) if n is below the function

there is a multiple of n that eventually upper bounds the function

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notation

$n^2 + n + 1 \in O(n^2)$

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big O: eventually upper bounded
Ω : eventually less than
$f_0(n) \geq f(n)$

Insertion sort

playing card sort: poker sorting cards as you go
find right place to insert
swap cards if is the right spot
easiest to put on right

```
def insertion_sort(A,N):
    for i ∈ 0,1,...,N-1:          # i is the sorting hand
        j ∈ i
        while j > 0 and A[j] < A[j-1]:
            swap
            j ∈ j-1
    return A
```



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while j check

sum of $[1,\dots,N] = \frac{n}{2}(n+1)$

inside while loop check
sum of $[0,\dots,N-1] = \frac{n}{2}(0+n-1)$

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average case running time
- expected value when input is chosen from a distribution

distribution: uniform over permutations of $[1,2,\dots,n-1,n]$

- weight all permutations evenly
- pick one permutation over n! permutations
- run it through insertion sort and count running time
- calculate expected value

- run and mean of expected value

Analysis

Outer Loop i:



P(only one swap) = $\frac{1}{i+1}$

there are i+1 slots.

Possible outcomes	Pos	Prob
0 swaps	i	$\frac{1}{i+1}$
1 swap	i-1	$\frac{1}{i+1}$
⋮	⋮	⋮
i swaps	0	$\frac{1}{i+1}$

equally likely i+1 outcomes

dominated by the number of swaps

Big O

$O(f(n))$ is the set of functions that are asymptotically upper bounded by a constant multiple of $f(n)$ without constant factors

Given $f(n) = n\sin(n) + n\cos(n)$

how do we know:
$f(n) \stackrel{?}{\in} O(n^2)$
$f(n) \stackrel{?}{\in} O(\sqrt{n})$

$O(f(n)) = \{ g(n) \mid \exists f(n), N_{f_0}, \forall n \geq N_{f_0}, 0 \leq g(n) \leq C_{f_0} f(n) \}$

placeholder variable ↑ (obligation)

Proving something is $\in O(f(n))$

Proof: $n+1 \in O(n)$

By the definition of $O(n)$, it suffices to show that there exists numbers C and N such that for every $n \geq N$, $0 \leq n+1 \leq Cn$.

Pick $C=2$ and $N=1$. Thus picking an arbitrary number $n \geq 1$, we needs to show $0 \leq n+1 \leq 2n$.

Since $n \geq 1$, we know that $0 \leq n$ and $0 \leq 1$. Thus, $0 \leq n+1$ is true. In addition, $1 \leq n$ can be manipulated by adding n to both sides $1+n \leq 2n$.

Therefore, both parts of the inequality are satisfied and we have shown $n+1 \in O(n)$.

Proof: $3n \in O(n^2)$

By the definition of $O(n^2)$, it suffices to show that there exists numbers C and N such that for all $n \geq N$, $0 \leq 3n \leq Cn^2$

Pick $C=3$ and $N=1$, Then for an arbitrary number $n \geq 1$, we need to show $0 \leq 3n \leq 3n^2$.

Since $n \geq 1$, $0 \leq 3n$ is true. Since addition, since $1 \leq n$. If we multiply by $3n$, which is a positive number, $3n \leq 3n^2$ holds true

Therefore, both parts of the inequality are satisfied and we have shown $3n \in O(n^2)$

Big Os and Equality (CS interpreted by a lawyer)

$O(n) + O(n^2) = O(n^2)$ does not make sense, can't add two sets

Definition: If an asymptotic set on the left hand side of equals has any other member.
Then exists a member of each asymptotic set on the r